

Akshar Patel

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EDUCATION

Indiana University Bloomington

Master of Science in Data Science (GPA: 4.0/4.0)

Aug 2025 – Present (Expected May 2027)

Bloomington, IN, USA

Charotar University of Science and Technology

Bachelor of Technology in Information Technology (GPA: 8.83/10)

Oct 2021 – May 2025

Anand, GJ, India

TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL (window functions, CTEs), Pinecone, Snowflake, PostgreSQL, MongoDB
- **ML & Data Libraries:** PyTorch, Hugging Face Transformers, scikit-learn, Pandas, NumPy, PySpark, Matplotlib
- **NLP & Generative AI:** RAG, multi-agent systems, LoRA fine-tuning, MCP (Model Context Protocol), LLM evaluation
- **Cloud & DevOps:** AWS (EC2, S3), Microsoft Azure, Docker, Git, GitHub Actions
- **Frameworks & Platforms:** FastAPI, Flask, REST APIs, Microsoft Copilot Studio, Power Automate, n8n

EXPERIENCE

AI Engineering Intern | Hitachi (HGAP) | Michigan City, IN

May 2026 – Present

- Architected a multi-agent AI assistant in Microsoft Copilot Studio that replaced manual document search, orchestrating 5 RAG agents via a supervisor-worker pattern with guardrails; deployed on Teams to 60 users across 5 departments.
- Integrated Snowflake through a managed MCP (Model Context Protocol) server, exposing Cortex Search and Cortex Analyst as agent tools for governed, real-time retrieval over structured and unstructured data.
- Built a human-in-the-loop escalation pipeline in Power Automate, routing unanswered queries to subject-matter experts through Teams adaptive cards and SharePoint with automated email notifications, resolving 50+ queries at a 3-hour average turnaround.

ML Engineer (Part-time) | Indiana University Bloomington | Bloomington, IN

Oct 2025 – Present

- Engineered a query-routing system for Indiana University's One.IU portal, mapping user queries to a closed catalog of 33 applications at 98.6% accuracy, eliminating hallucinated routes by design.
- Implemented a hybrid routing pipeline combining embedding search (all-MiniLM-L6-v2), fuzzy matching, and an LLM tiebreaker, reducing inference cost through a 3-stage validation cascade.
- Deployed a REST API on Azure App Service that fetches and serves live IU dining data; adopted by the ChatAIU team as an agent tool, closing a live-data gap in the university's AI assistant.
- Developed a Flask-based LLM autograding platform integrated with Canvas LMS via OAuth2, automatically retrieving submissions and grading against faculty rubrics using a schema-validated LLM agent with faculty review before Excel export.

Research Assistant | Indiana University Bloomington | Bloomington, IN

Jan 2026 – May 2026

- Fine-tuned RoBERTa-base/large with domain-adaptive pretraining and LoRA on Big Red 200 HPC GPUs for mental-health NLP classification on 700K+ samples, improving macro-F1 by 9 points over classical ML baselines.
- Trained CNN and Vision Transformer models (ResNet, EfficientNet, ViT) in PyTorch on a 15K+ image dataset, applying focal loss for class imbalance and staged fine-tuning to achieve 0.82 ROC-AUC.

Data Science Intern (Remote) | Scale Capacity | Los Angeles, CA

Jun 2024 – Aug 2024

- Benchmarked Naive Bayes, Logistic Regression, SVM, Random Forest, and ensemble classifiers on 50K+ TF-IDF-vectorized records for spam detection (SQL, Pandas, PySpark), achieving 97% precision and 96% recall on imbalanced data.

PROJECTS

EduPilot – Multi-Agent, Source-Grounded RAG System

[GitHub](#)

- Orchestrated a 7-stage multi-agent RAG pipeline (FastAPI, Pinecone, Groq Llama 3.3 70B) with hybrid semantic and BM25 retrieval fused via Reciprocal Rank Fusion, inline source citations, and LLM-as-judge verification.
- Evaluated the system with a 50-case, 8-metric suite spanning faithfulness, citation accuracy, and adversarial hallucination tests, quantifying a 3.7x citation-accuracy gap between 70B and 8B models.

IntelliSphere – Domain-Specific RAG Conversational AI

[GitHub](#)

- Created a domain-specific RAG assistant for healthcare and legal document QA (Flask, LangChain, FAISS, Gemini) with recursive chunking, metadata-tagged indexing, and MongoDB-backed multi-session conversation memory.
- Containerized the stack with Docker and shipped it via GitHub Actions CI/CD, building images to Amazon ECR and auto-deploying to EC2 through a self-hosted runner, with FAISS indexes synced from S3 at startup.